

AI Critique

AI was useful for accelerating the initial Playwright structure, converting manual cases into data-driven tests, and identifying reusable patterns such as Page Objects and API-based setup. However, its first output was incomplete in several important ways. For registration testing, AI initially used the specification-valid password in every case. Because the SUT contains a flawed regular expression that requires whitespace instead of a special character, the password check blocked invalid-email tests before the email behavior was reached. Those tests could therefore fail for the wrong reason. Human review corrected this by isolating one variable at a time and using data that deliberately passes the SUT's flawed password gate when testing email validation.

AI also underestimated shared-state problems. Running three browser projects can create duplicate users, orders, and categories. Unique per-worker data and API-created prerequisites were added after review. Fixed or implicit waits were replaced with response-based synchronization, and negative API tests were strengthened to assert exact HTTP status codes. Another gap was environmental: an `import.meta`-based JSON path failed under the actual module configuration, so it was replaced with a working-directory path.

The main lesson is that AI-generated automation is only a draft. A plausible script is not necessarily a valid test. Effective collaboration requires checking whether each assertion observes the intended behavior, whether the test can pass or fail for an unrelated reason, and whether repeated execution is deterministic. AI is strongest at producing structure quickly; the student must supply requirement judgment, failure diagnosis, and responsibility for the final evidence.